MAST20009 Vector Calculus

Problem Sheet 2: Space Curves and Vector Fields

Worked Solutions for selected questions

Q29. Rolling Wheel. Solution:

$$\begin{aligned}
\boldsymbol{c}(t) &= \left(vt - R\sin\left(\frac{vt}{R}\right), R - R\cos\left(\frac{vt}{R}\right)\right) \\
\boldsymbol{v}(t) &= \frac{dc}{dt} \\
\boldsymbol{v}(t) &= \left(v - v\cos\left(\frac{vt}{R}\right), v\sin\left(\frac{vt}{R}\right)\right)
\end{aligned}$$

The velocity vector is horizontal when the y-component of velocity is zero. We assume that $v \neq 0$. Thus

$$v\sin\left(\frac{vt}{R}\right) = 0 \Rightarrow \quad \frac{vt}{R} = n\pi, \text{ for some } n \in \mathbb{Z} \Rightarrow t = \frac{n\pi R}{v}.$$

So the velocity vector is horizontal when $t = \frac{n\pi R}{v}$, $n \in \mathbb{Z}$. Now the speed is given by

$$\begin{aligned}
\text{speed} = \|\boldsymbol{v}\| &= \sqrt{\left(v - v\cos\left(\frac{vt}{R}\right)\right)^2 + \left(v\sin\left(\frac{vt}{R}\right)\right)^2} \\
&= \sqrt{v^2 - 2v^2\cos\left(\frac{vt}{R}\right) + v^2\cos^2\left(\frac{vt}{R}\right) + v^2\sin^2\left(\frac{vt}{R}\right)} \\
&= \sqrt{2v^2 - 2v^2\cos\left(\frac{vt}{R}\right)} = 2v\left|\sin\left(\frac{vt}{2R}\right)\right| \qquad \text{by double angle formula}
\end{aligned}$$

Therefore the speed at $t = \frac{n\pi R}{v}$ is

$$\text{speed} = 2v\left|\sin\left(\frac{n\pi}{2}\right)\right| = \begin{cases} 0, & \text{n is even,} \\ 2v, & \text{n is odd .} \end{cases}$$

Q31 (a). Solution: $\boldsymbol{F}$ is a differentiable function of t

$$\begin{aligned}
\frac{d}{dt}\left(\boldsymbol{F}\cdot\frac{d\boldsymbol{F}}{dt}\times\frac{d^2\boldsymbol{F}}{dt^2}\right) &= \frac{d\boldsymbol{F}}{dt}\cdot\left(\frac{d\boldsymbol{F}}{dt}\times\frac{d^2\boldsymbol{F}}{dt^2}\right) + \boldsymbol{F}\cdot\frac{d}{dt}\left(\frac{d\boldsymbol{F}}{dt}\times\frac{d^2\boldsymbol{F}}{dt^2}\right) \\
&= 0 + \boldsymbol{F}\cdot\left(\frac{d^2\boldsymbol{F}}{dt^2}\times\frac{d^2\boldsymbol{F}}{dt^2} + \frac{d\boldsymbol{F}}{dt}\times\frac{d^3\boldsymbol{F}}{dt^3}\right) \qquad \text{by using } \boldsymbol{a}\cdot(\boldsymbol{a}\times\boldsymbol{b}) = \boldsymbol{0} \\
&= \boldsymbol{F}\cdot\left(\boldsymbol{0} + \frac{d\boldsymbol{F}}{dt}\times\frac{d^3\boldsymbol{F}}{dt^3}\right) \qquad \text{since } \boldsymbol{a}\times\boldsymbol{a} = \boldsymbol{0} \\
&= \boldsymbol{F}\cdot\left(\frac{d\boldsymbol{F}}{dt}\times\frac{d^3\boldsymbol{F}}{dt^3}\right)
\end{aligned}$$

Q34 (a). Curve Properties. Solution:

Method 1: We know that

$$\boldsymbol{B}\cdot\boldsymbol{B}=1\,.$$

Differentiate both sides with respect to t

$$\begin{aligned}
&\frac{d}{dt}\left(\boldsymbol{B}\cdot\boldsymbol{B}\right)=\frac{d}{dt}(1)\\
&\implies \boldsymbol{B}\cdot\frac{d\boldsymbol{B}}{dt}+\frac{d\boldsymbol{B}}{dt}\cdot\boldsymbol{B}=0 \qquad \text{by product rule of differentiation}\\
&\implies 2\,\boldsymbol{B}\cdot\frac{d\boldsymbol{B}}{dt}=0 \qquad \text{by commutative law of dot product}\\
&\implies \boldsymbol{B}\cdot\frac{d\boldsymbol{B}}{dt}=0\,.
\end{aligned}$$

Method 2: We know that $\boldsymbol{B}=\boldsymbol{T}\times\boldsymbol{N}$. Thus

$$\begin{aligned}
&\frac{d\boldsymbol{B}}{dt}=\frac{d}{dt}\left(\boldsymbol{T}\times\boldsymbol{N}\right)=\boldsymbol{T}\times\frac{d\boldsymbol{N}}{dt}+\frac{d\boldsymbol{T}}{dt}\times\boldsymbol{N}=\boldsymbol{T}\times\frac{d\boldsymbol{N}}{dt}\ \text{since}\ \boldsymbol{N}=\frac{1}{\|\frac{d\boldsymbol{T}}{dt}\|}\frac{d\boldsymbol{T}}{dt}\\
\Rightarrow&\frac{d\boldsymbol{B}}{dt}\cdot\boldsymbol{B}=\left(\boldsymbol{T}\times\frac{d\boldsymbol{N}}{dt}\right)\cdot\boldsymbol{B}=\boldsymbol{B}\cdot\left(\boldsymbol{T}\times\frac{d\boldsymbol{N}}{dt}\right)\\
&=\frac{d\boldsymbol{N}}{dt}\cdot\left(\boldsymbol{B}\times\boldsymbol{T}\right) \qquad \text{since}\ \boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c})=\boldsymbol{c}\cdot(\boldsymbol{a}\times\boldsymbol{b})=\boldsymbol{b}\cdot(\boldsymbol{c}\times\boldsymbol{a})\\
&=\frac{d\boldsymbol{N}}{dt}\cdot\left((\boldsymbol{T}\times\boldsymbol{N})\times\boldsymbol{T}\right) \qquad \text{since}\ \boldsymbol{B}=\boldsymbol{T}\times\boldsymbol{N},\\
&=-\frac{d\boldsymbol{N}}{dt}\cdot\left(\boldsymbol{T}\times(\boldsymbol{T}\times\boldsymbol{N})\right) \qquad \text{since}\ \boldsymbol{a}\times\boldsymbol{b}=-\boldsymbol{b}\times\boldsymbol{a},\\
&=-\frac{d\boldsymbol{N}}{dt}\cdot\left((\boldsymbol{T}\cdot\boldsymbol{N})\,\boldsymbol{T}-(\boldsymbol{T}\cdot\boldsymbol{T})\,\boldsymbol{N}\right) \qquad \text{since}\ \boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c})=(\boldsymbol{a}\cdot\boldsymbol{c})\boldsymbol{b}-(\boldsymbol{a}\cdot\boldsymbol{b})\boldsymbol{c}))\\
&=-\left(\boldsymbol{T}\cdot\boldsymbol{N}\right)\left(\frac{d\boldsymbol{N}}{dt}\cdot\boldsymbol{T}\right)+\left(\boldsymbol{T}\cdot\boldsymbol{T}\right)\left(\boldsymbol{N}\cdot\frac{d\boldsymbol{N}}{dt}\right) \qquad \text{by distributive law}\\
&=-\left(0\right)\left(\frac{d\boldsymbol{N}}{dt}\cdot\boldsymbol{T}\right)+\left(\boldsymbol{T}\cdot\boldsymbol{T}\right)(0) \qquad \text{as}\ \boldsymbol{T}\perp\boldsymbol{N}\ \text{and}\ \boldsymbol{N}\perp\frac{d\boldsymbol{N}}{dt}\\
\Rightarrow&\frac{d\boldsymbol{B}}{dt}\cdot\boldsymbol{B}=0
\end{aligned}$$

Q36. Parametric Surface 1.

Solution:

$$x=2u,\quad y=u^2+v,\quad z=v^2$$

(a) Tangent vectors to surface are: $\boldsymbol{T}_u=(2,2u,0),\quad \boldsymbol{T}_v=(0,1,2v)$.
A normal vector to surface is

$$\boldsymbol{n}=\boldsymbol{T}_u\times\boldsymbol{T}_v=\begin{vmatrix}\mathbf{i} & \mathbf{j} & \mathbf{k}\\ 2 & 2u & 0\\ 0 & 1 & 2v\end{vmatrix}=\mathbf{i}\,(4uv)-\ \mathbf{j}\,(4v)+\mathbf{k}\,(2)=(4uv,-4v,2)$$

(b) Since $\boldsymbol{n} \neq \mathbf{0}$ for all u, v, the surface is smooth for all u and v.
(c) If $(x_0, y_0, z_0) = (0, 1, 1)$

$$x_0 = 2u_0 \Rightarrow u_0 = 0, \tag{1}$$
$$y_0 = {u_0}^2 + v_0 \Rightarrow v_0 = 1, \tag{2}$$
$$z_0 = {v_0}^2 \Rightarrow {v_0}^2 = 1 \Rightarrow v_0 = \pm 1 \tag{3}$$

(1), (2) and (3) give $u_0 = 0, v_0 = 1$. Normal at $(u_0, v_0) = (0, 1)$ is

$$\boldsymbol{n}(u_0, v_0) = (0, -4, 2)$$

A tangent plane to surface at $(0, 1, 1)$ is

$$\begin{aligned}(x - x_0, y - y_0, z - z_0) \cdot \boldsymbol{n}(u_0, v_0) &= 0\\ (x, y - 1, z - 1) \cdot (0, -4, 2) &= 0\\ -2y + z + 1 &= 0\end{aligned}$$

Q43 (b). Solution:

$$\boldsymbol{F} = (x, -y)$$

For the flow lines:

$$\boldsymbol{c}' = \boldsymbol{F}(\boldsymbol{c}(t)) \qquad \text{so } \left(\frac{dx}{dt}, \frac{dy}{dt}\right) = (x, -y) \implies \frac{dx}{dt} = x, \quad \frac{dy}{dt} = -y$$

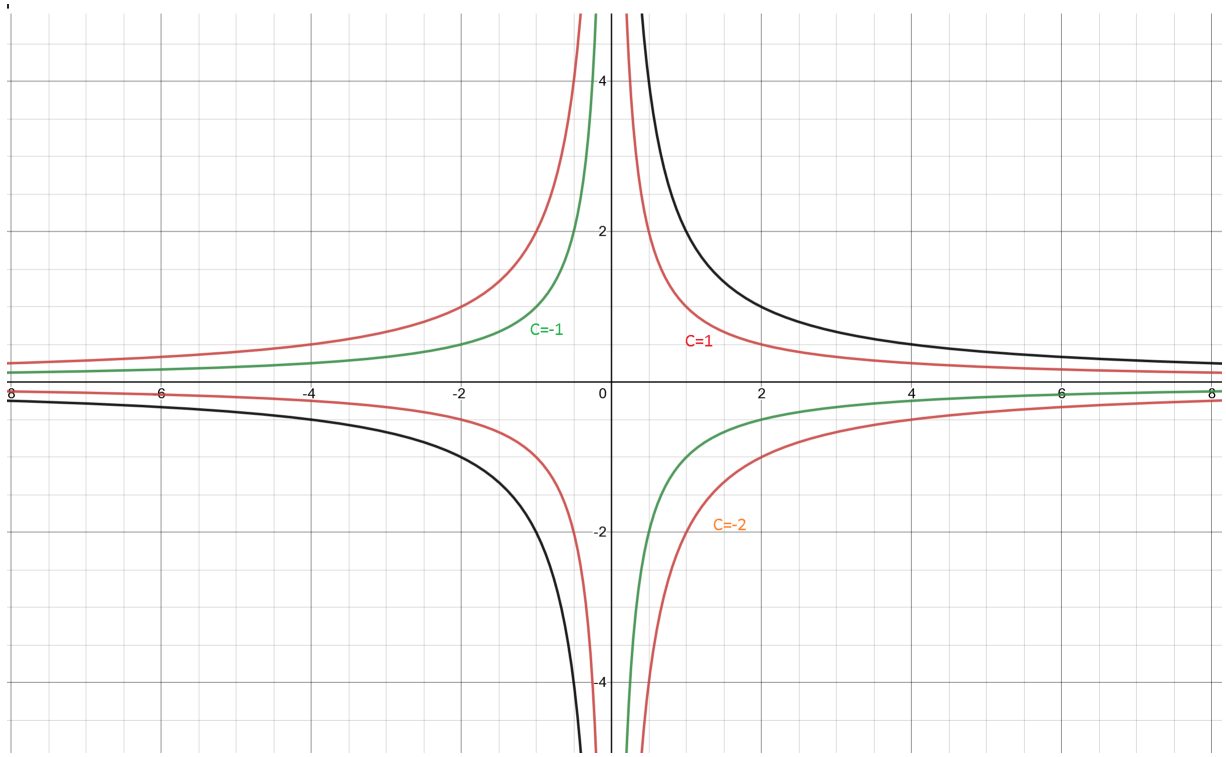
$$\frac{dy}{dx} = \frac{dy}{dt}\frac{dt}{dx} \Rightarrow \quad \frac{dy}{dx} = -\frac{y}{x}$$

$$\int \frac{1}{y}\frac{dy}{dx}\,dx = -\int \frac{1}{x}\,dx \qquad \text{by separating variables and integrate with respect to } x \text{ on both sides}$$

$$\log|y| = -\log|x| + c, \qquad \text{where } c \in \mathbb{R} \text{ is an arbitrary constant}$$

$$\log|xy| = c \implies xy = C$$

$$y = \frac{C}{x} \qquad \text{represents the flow lines (a family of hyperbola) for } \boldsymbol{F}.$$



Q45 (c). Solution:

$$\boldsymbol{V}(x,y,z) = \sin(xy)\mathbf{i} - \cos(x^2y)\mathbf{j}$$
$$\operatorname{div}\boldsymbol{V} = \boldsymbol{\nabla}\cdot\boldsymbol{V} = \frac{\partial}{\partial x}\sin(xy) - \frac{\partial}{\partial y}\cos(x^2y) = y\cos(xy) + x^2\sin(x^2y)$$

Q46 (c) Solution:

$$\boldsymbol{V}(x,y,z) = (x^2+y^2+z^2)(3\mathbf{i}+4\mathbf{j}+5\mathbf{k})$$

$$\mathbf{Curl}\,\boldsymbol{V} = \boldsymbol{\nabla}\times\boldsymbol{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 3(x^2+y^2+z^2) & 4(x^2+y^2+z^2) & 5(x^2+y^2+z^2) \end{vmatrix}$$

$$\begin{aligned} &= \mathbf{i}\left(\frac{\partial}{\partial y}(5(x^2+y^2+z^2)) - \frac{\partial}{\partial z}(4(x^2+y^2+z^2))\right) \\ &\quad - \mathbf{j}\left(\frac{\partial}{\partial x}(5(x^2+y^2+z^2)) - \frac{\partial}{\partial z}(3(x^2+y^2+z^2)\right) \\ &\quad + \mathbf{k}\left(\frac{\partial}{\partial x}(4(x^2+y^2+z^2)) - \frac{\partial}{\partial y}(3(x^2+y^2+z^2)\right) \\ &= (10y-8z)\mathbf{i} + (6z-10x)\mathbf{j} + (8x-6y)\mathbf{k} \end{aligned}$$

Q49 (a)

Solution: $\boldsymbol{F}(x, y, z) = 2x\mathbf{i} + 3y\mathbf{j} + 4z\mathbf{k}$

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ 2x & 3y & 4z \end{vmatrix} = \mathbf{i}\,(0-0) - \ \mathbf{j}\,(0-0) + \ \mathbf{k}\,(0-0) = \mathbf{0}.$$

Since $\nabla \times \mathbf{F} = \mathbf{0}$, $\boldsymbol{F}$ is an irrotational vector field.
If $\boldsymbol{F} = \nabla\phi$ then

$$\left(\frac{\partial\phi}{\partial x}, \frac{\partial\phi}{\partial y}, \frac{\partial\phi}{\partial z}\right) = (2x, 3y, 4z)$$

Equate components:

$$\frac{\partial\phi}{\partial x} = 2x$$
$$\frac{\partial\phi}{\partial y} = 3y$$
$$\frac{\partial\phi}{\partial z} = 4z$$

Integrating gives:

$$\phi = \int 2x\,dx = x^2 + C_1(y, z), \tag{4}$$
$$\phi = \int 3y\,dy = \frac{3}{2}y^2 + C_2(x, z), \tag{5}$$
$$\phi = \int 4z\,dz = 2z^2 + C_3(x, y). \tag{6}$$

Comparing Eqs(4-6) gives the scalar potentials

$$\phi = x^2 + \frac{3}{2}y^2 + 2z^2 + C,$$

where C is a constant.